HARD · PROBLEM-SOLVING AND DATA ANALYSIS

Mixed

04

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

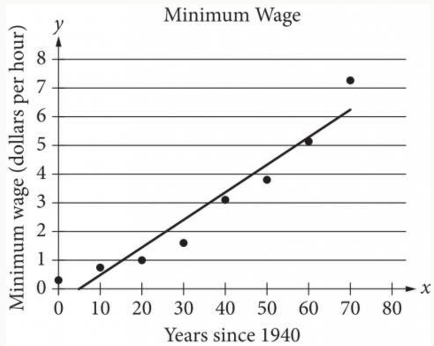
Q1 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

The number w is 110% greater than the number z. The number z is 55% less than 50. What is the value of w?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



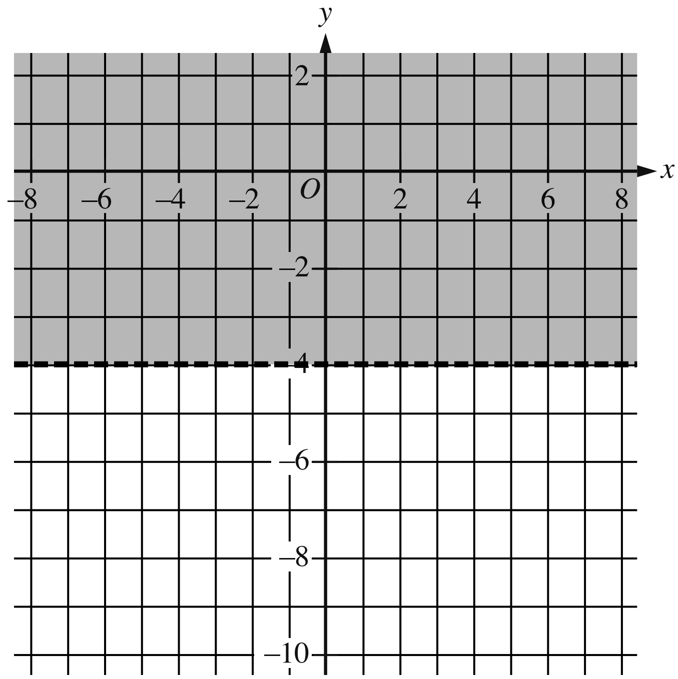
The scatterplot above shows the federal-mandated minimum wage every 10 years between 1940 and 2010. A line of best fit is shown, and its equation is $y = 0.096x - 0.488$. What does the line of best fit predict about the increase in the minimum wage over the 70-year period?

A. Each year between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.

B. Each year between 1940 and 2010, the average increase in minimum wage was 0.49 dollars.

C. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.

D. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.488 dollars.



- The boundary of the inequality is a dashed line.
  - The line is horizontal.
  - The line passes through the following points:
    - (negative 3 comma negative 4)
    - (0 comma negative 4)
    - (3 comma negative 4)
- The area above the boundary is shaded.

The shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. What is the value of r?

A. 15

B. 4

C. -4

D. -15

Q4 ALGEBRA

$$-12x + 14y = 36$$

$$-6x + 7y = -18$$

How many solutions does the given system of equations have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Q5

ALGEBRA

x	fx
-4	0
$-\frac{19}{5}$	1
$-\frac{18}{5}$	2

For the linear function f, the table shows three values of x and their corresponding values of fx. If $hx = fx - 13$, which equation defines h?

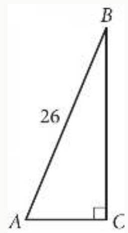
A. $hx = 5x - 4$

B. $hx = 5x + 7$

C. $hx = 5x + 9$

D. $hx = 5x + 20$

Q6 GRID-IN GEOMETRY AND TRIGONOMETRY



Triangle ABC above is a right triangle, and $\sin(B) = \frac{5}{13}$. What is the length of side $\overline{BC}$?

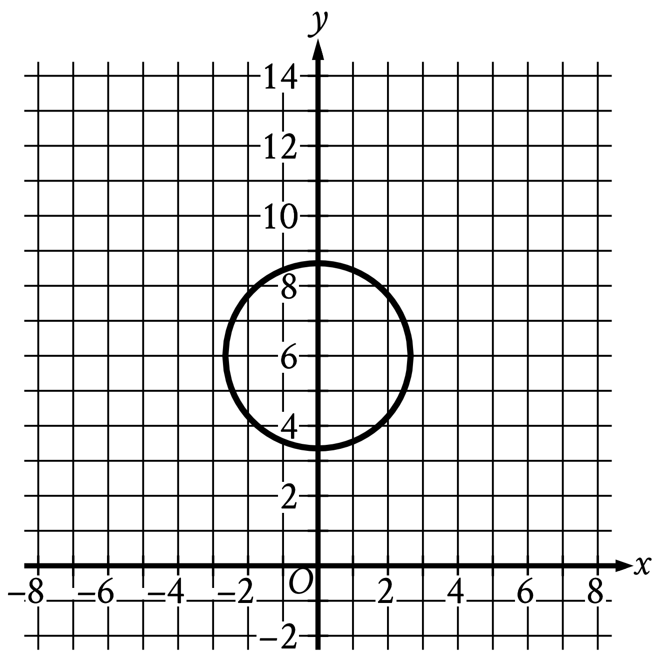
STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

GRID-IN GEOMETRY AND TRIGONOMETRY



- The circle passes through the following approximate points:
  - (0 comma 8.6)
  - (2.6 comma 6)
  - (0 comma 3.4)
  - (negative 2.6 comma 6)

Circle A shown is defined by the equation $x^2 + y - 6^2 = 7$. Circle B (not shown) has the same radius but is translated 96 units to the right. If the equation of circle B is $x - h^2 + y - k^2 = a$, where h, k, and a are constants, what is the value of $4a$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8 ADVANCED MATH

$$D = T - \frac{9}{25}(100 - H)$$

The formula above can be used to approximate the dew point D, in degrees Fahrenheit, given the temperature T, in degrees Fahrenheit, and the relative humidity of H percent, where $H > 50$. Which of the following expresses the relative humidity in terms of the temperature and the dew point?

A. $H = \frac{25}{9}(D - T) + 100$

B. $H = \frac{25}{9}(D - T) - 100$

C. $H = \frac{25}{9}(D + T) + 100$

D. $H = \frac{25}{9}(D + T) - 100$

Q9

ADVANCED MATH

$$57x^2 + \left(57b + a\right)x + ab = 0$$

In the given equation, a and b are positive constants. The product of the solutions to the given equation is kab, where k is a constant. What is the value of k?

A. $\frac{1}{57}$

B. $\frac{1}{19}$

C. 1

D. 57

Q10 GRID-IN ADVANCED MATH

In the xy-plane, the graph of $y=3x^2-14x$ intersects the graph of $y=x$ at the points $(0, 0)$ and (a, a). What is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.